Personal Statement
Applicant: Mubashir Ali
Program: Bachelor’s degree in bioinformatics / Biological Science
Zhejiang University, China
Date: 12/5/2025

From an early age, I have been captivated by the power of technology to decode and solve complex problems in the natural world. This fascination led me to pursue a dual path in bioinformatics and computer science, bridging computational methods with cutting-edge biological research. My academic journey at Quaid-i-Azam University in Bioinformatics and the University of the People in Computer Science has strengthened my foundation in algorithms, machine learning, and computational genomics, preparing me to contribute meaningfully to the field of genome engineering.

My research interests focus on leveraging Artificial Intelligence and Machine Learning to optimize CRISPR-Cas9 gene-editing technologies. Projects such as GeneFix AI, which predicts off-target effects and enhances precision in gene-editing experiments, and Bio Data Hub, a VS Code extension for streamlined analysis of biological datasets, reflect my commitment to developing computational tools that accelerate real-world applications in healthcare and genomics. Beyond research, I have actively applied my skills as a founder of Code with Bismillah, mentoring students, building AI-driven platforms, and creating educational content to make complex technologies accessible.

I am particularly drawn to Zhejiang University for its distinguished faculty, interdisciplinary programs, and strong focus on bioinformatics and computational biology. The university’s resources and collaborative environment will allow me to deepen my expertise in AI-driven genomics, develop robust predictive models for disease diagnosis, and refine computational pipelines for genome analysis. My long-term goal is to design innovative tools that enable safer, faster, and more accurate genome editing, ultimately supporting advances in personalized medicine and biotechnology.

By joining Zhejiang University, I aim to merge my computational expertise with biological insights, contributing to transformative research in bioinformatics. I am eager to collaborate with leading researchers, engage with diverse perspectives, and translate AI innovations into practical solutions for healthcare and life sciences. This program represents the ideal environment to hone my skills, broaden my knowledge, and make a meaningful impact on global genomics research.